C9orf72-Nature2Agent

An autonomous AI scientist built from the C9orf72 ALS-autoimmunity study

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Abstract

This report describes a multi-agent system that converts the *Nature* study on C9orf72 autoimmunity in amyotrophic lateral sclerosis (ALS) [1] into an interactive, executable agent. The design choice that defines the system is a separation between a deterministic computational core, which produces every reported number, and a reasoning layer, which chooses what to ask. The core is exposed as validated tools behind a Model Context Protocol (MCP) server, and a five-layer agent ecosystem uses those tools to explain the design, reproduce the analyses and figures, interrogate the peptide and HLA data, and pose and test questions beyond the paper. Because every result is computed, the grounding rate is one by construction and the discovery loop reproduces without credentials. In the committed configuration the questions come from a curated bank executed deterministically; a language-model backend can drive the same agents live when an API key is present. Reproduced p-values match the authors' reported values to within $|\Delta p| = 0.00131$ with full agreement of the significance calls. Beyond reproduction, the system reports two genuinely new biological observations, that epitope immunogenicity is spatially clustered along the protein sequence (permutation $p = 2.00 \times 10^{-4}$) and that the ALS IL-5 bias is shared with TDP-43 while the IL-10 bias is specific to C9orf72, together with a set of robustness audits of the paper's own effects. It evaluates both the science (reproducibility, robustness, novelty, falsifiability) and itself (grounding, tool-use accuracy, determinism, rejection of planted false hypotheses). A patient-stratified antigen-specific therapeutic hypothesis, with an explicit falsification experiment, is included as an illustrative downstream use.

1 Introduction

Amyotrophic lateral sclerosis is a fatal neurodegenerative disease defined by the progressive loss of motor neurons. Neuroinflammation accompanies the degeneration, with T-cell infiltration and microglial activation in affected tissue, and T cells are thought to contribute to the pace of neuronal loss [2, 3]. A specific antigenic target for this response had not been reported until Michaelis et al. identified C9orf72 as a target of ALS-associated T-cell autoreactivity and mapped the recognised epitopes [1]. Their central observations are that C9orf72-specific responses are increased in ALS, that the responses are mediated mainly by $CD4^+$ T cells, that they are relatively enriched for IL-5 and IL-10 over IFNγ, and that IL-10-mediated responses are associated with longer predicted survival.

The objective is to ask whether an AI system can use the C9orf72 autoimmune mechanism as a starting point and move from observation to mechanism to hypothesis to a falsifiable therapeutic proposal, while remaining honest about what the data support. This is a question about reasoning over the study, not a re-run of it. The work follows the Paper2Agent paradigm, which turns a

paper and its computational resources into an executable agent served over MCP, so the paper becomes a resource an agent can run [4].

The difficulty in automating this is reliability. A system that reads a paper and proposes hypotheses can produce fluent text that states numbers the data do not support, and in a scientific setting such a claim is worse than silence, because it is costly to check and easy to believe. The position taken here is that the way to earn trust is to make every reported number the output of a computation that anyone can rerun, and to make the system's honesty a measured quantity. That choice shapes the whole design, from the shared tool core to the grounding and false-hypothesis metrics, and it is the reason the system is cautious where it could be impressive. This report is one of a set of related projects that share a dataset and are meant to interoperate, so the emphasis on tidy, computed, auditable outputs also serves the larger goal of letting such systems be combined without compounding each other's errors.

Contributions. The central contribution is methodological: a design in which every reported number is produced by a shared deterministic core, so reproduction, robustness checks and hypothesis-testing are all grounded and auditable, and the system's honesty can be measured (grounding, determinism, and the rejection of planted false hypotheses). Two further contributions follow. The first is building the MCP tool layer and the analyses from the raw open data for a paper that has no public analysis codebase, which is where this departs from wrapping an existing tutorial. The second is scientific: alongside reproducing the study, the system reports two new observations and a set of robustness audits, and frames one falsifiable downstream hypothesis. All code was written with an AI coding agent, inspected, and covered by an automated test suite.

The pipeline the system is meant to demonstrate. The aim is to show that a multi-agent system can move along a chain from an observation to a mechanism, from a mechanism to a hypothesis, from a hypothesis to a computational result, and from a result to a therapeutic candidate with a falsification experiment attached. Each link in that chain is a concrete artefact in this project: the observations are the reproductions, the mechanism is the IL-10-biased C9orf72-specific response, the hypotheses are the generated questions, the computational results are the verified observations, the candidate is the antigen-specific regulatory strategy, and the falsification is the experiment the designer agent specifies. The value of making the chain explicit is that every step can be checked, and a weak step stays visible for inspection.

2 Related work

Two lines of work frame this project. The first is the conversion of scientific artefacts into executable agents. Paper2Agent automates the construction of an MCP server from a paper and its codebase, with agents that locate the code, set up the environment, extract tools from tutorials and validate them before assembly [4]. The Michaelis study is an experimental immunology paper without a public analysis codebase, so the present work builds the computational infrastructure directly from the data, then exposes it in the same MCP tool shape. The second line is multi-agent hypothesis generation, where separate agents propose, critique and debate hypotheses and a coordinating agent integrates them. The discovery loop here follows that division of labour, with explicit verification, replication, contradiction and falsification agents, and a principal-investigator agent that integrates without the authority to overrule evidence.

A recurring concern in both lines is the reliability of language-model output in a scientific setting, where a fluent but unsupported claim is worse than no claim. The design response taken here is to separate a deterministic computational core, which produces every number, from the language

layer, which only frames and routes. In the committed configuration the framing layer is rule-based and reads from the computed results, so a figure in the text is always a figure that was computed. The evaluation in Section 11 measures this directly through a grounding check and a planted-false-hypothesis test.

2.1 Positioning

It helps to be explicit about what this work adds, because the space of paper-to-agent and AI-scientist systems is active. Against Paper2Agent, the difference is the starting material. Paper2Agent agentifies a paper by locating and wrapping its existing code and tutorials; the study used here is an experimental immunology paper with open data but no analysis codebase, so there is nothing to wrap, and the tool layer is built from the raw data. The contribution there is showing that the same output shape, an MCP server of validated tools, can be reached for a data-only paper. Against language-model "AI scientist" and co-scientist systems, which generate and debate hypotheses with a model in the loop, the difference is where the trust sits. Those systems are strong at proposing and weak at guaranteeing that a stated number is real. This system inverts the emphasis: the proposing is modest, bounded by a question bank in the committed configuration, but every number is computed by a shared core and the honesty of the system is measured. The two designs are complementary, and the natural combination, a language model proposing over this grounded core, is the clearest next step.

3 Background and data

3.1 C9orf72 genetics and biology

A GGGGCC hexanucleotide repeat expansion in the first intron of *C9orf72* is the most common genetic cause of amyotrophic lateral sclerosis and frontotemporal dementia, and it accounts for roughly one in ten cases of ALS overall and a larger share of familial disease. Unaffected alleles carry a small number of repeats, while pathogenic alleles carry hundreds to thousands. Three mechanisms are thought to act together: a loss of C9orf72 protein, a gain of function from the repeat RNA, which forms foci that sequester RNA-binding proteins, and repeat-associated non-ATG translation, which produces dipeptide repeat proteins. The protein itself acts in autophagy and membrane trafficking as a guanine-nucleotide exchange factor for Rab GTPases, and it also shapes myeloid and lymphoid immune function, so a reduction in its level has consequences for both neuronal maintenance and immune regulation. This dual role is part of what makes an autoimmune response against the protein plausible.

3.2 Immune involvement in ALS

Neuroinflammation accompanies the degeneration of motor neurons. Microglial activation and infiltrating monocytes are seen in ALS model tissue [5], lymphocyte infiltration and glial activation are described across human and model studies [2], and the cellular niche around motor neurons is altered [3]. The adaptive arm plays an active role. Regulatory T cells are neuroprotective in ALS, and their number and suppressive function are reduced in rapidly progressing patients, so a weaker regulatory compartment is associated with faster progression [6]. Regulatory T-cell dysfunction is also documented in ALS [7], and the extension of survival by regulatory T cells in disease models has motivated Treg-directed therapy [8]. The cytokine context matters as well. IL-10, the principal cytokine of regulatory T cells, is linked to neuroprotection through the dampening of microglial activation across neurodegeneration models [9–11]. What had been

missing was a defined self-antigen that the T-cell response targets. The study used here fills that gap by identifying the C9orf72 protein as a target of ALS-associated T-cell autoreactivity and by mapping the recognised epitopes.

3.3 Antigen-specific immunotherapy

The therapeutic interest in this mechanism follows from the regulatory angle. Broad immunosuppression is a blunt instrument in a disease where part of the immune response is protective, so the field has moved toward restoring or expanding the regulatory compartment. Autologous regulatory T-cell approaches have been tested in ALS [12], and antigen-specific tolerance is an active strategy in autoimmune disease more generally [13]. The evidence is mixed, because a recent polyclonal regulatory T-cell trial in a different autoimmune indication was negative, which tempers expectations [14]. A defined autoantigen changes the design space, because it opens the door to an antigen-specific, not a polyclonal intervention, and the data analysed here point to which antigen and which patients such an intervention might target.

3.4 The study and its data

The study examined peripheral blood mononuclear cells from 40 people with ALS and 28 healthy controls, with additional Parkinson's disease (PD) and Alzheimer's disease (AD) cohorts and a validation cohort. Antigen-specific responses were measured by IFNγ, IL-5 and IL-10 FluoroSpot after a two-week expansion, against pools of C9orf72, TDP-43 and SOD1 peptides and an EBV control pool. Immunophenotyping used spectral flow cytometry, and cytokine-producing cells were characterised by intracellular cytokine staining. HLA class II binding was predicted with IEDB NetMHCIIpan-4.1 EL [15, 16], and HLA restrictions were inferred with RATE.

All primary data are open. The flow-cytometry files and FlowJo workspaces are deposited on Zenodo (CC-BY 4.0) across one spectral and four intracellular-cytokine-staining records; the full text, the figure source data and the supplementary tables are open access on PubMed Central (PMC12657233). The system fetches these with checksum verification and records provenance for each file. From them it builds fourteen tidy tables, including the complete 100-peptide screen, the per-donor HLA typing of 23 donors, and the per-donor source data behind every figure.

3.5 Cohorts and assays

The response data come from several assays with different readouts, which the reconstruction keeps distinct. The cytokine response screen is a FluoroSpot assay. Thawed peripheral blood mononuclear cells are expanded for two weeks against a peptide pool, then restimulated, and the number of cells secreting IFNγ, IL-5 or IL-10 is counted as spot-forming units. This is the assay behind the total and per-cytokine response magnitudes and the per-donor values used for the main ALS-versus-HC comparisons. Intracellular cytokine staining on a conventional cytometer characterises which T-cell subsets produce the cytokines, and spectral cytometry on a high-parameter instrument profiles the broad T-helper compartment. The antigens are peptide pools for C9orf72, for the ALS proteins TDP-43 and SOD1, and for Epstein-Barr virus as a non-self control. The C9orf72 library tiles the 481-residue protein with 100 overlapping 15-mer peptides stepped by five residues, and it includes five phosphorylated variants at serine or threonine positions. Sample sizes differ by assay, so the reconstruction carries the group counts with each table, so no analysis assumes the full cohort throughout; the FluoroSpot screen used about 28 donors per group, within the larger 40 ALS and 28 control cohort.

The clinical variables attached to the donors are what make the survival and mutation analyses

possible, and they also bound them. The study records, for the relevant donors, the C9orf72 mutation status, a predicted survival time, and functional-rating and demographic variables, and the reconstruction carries the per-donor values behind the clinical figure so these can be related to the response magnitudes. The mutation and survival strata are small once the cohort is split, which is the reason those effects are treated as marginal throughout this report, and it is also why the system reports the stratified therapeutic reasoning as a hypothesis to test rather than a conclusion. The 23 donors with full HLA typing are a subset again, so the HLA-restriction results speak to those donors and to the alleles they carry, and the reconstruction keeps the donor identifiers so a reader can see exactly which donors enter each analysis. Keeping the counts and identifiers visible at every step is a small discipline that prevents the most common error in reusing someone else's data, which is to assume a single cohort where the original used several overlapping ones.

3.6 Machine-readable reconstruction

The source-data workbook stores each figure as several side-by-side panels, some of which reuse panel letters and measure names. A column-centric parser forward-fills the header rows and separates panels by column gaps, which prevents two different panels from being merged. The parse is cross-checked against the paper: epitope number 14 resolves to residues 56–70 (`DGEITFLANHTLNGE`), recognised by 7 of 22 donors with an average magnitude of 1216, which matches both the manuscript's Table 1 and the supplementary screen. The embedded C9orf72 sequence (UniProt Q96LT7, 481 residues) places all 95 non-phosphorylated peptides at their stated residue windows without error.

3.7 Acquisition and provenance

Acquisition is scripted and verified. The four intracellular-cytokine-staining Zenodo records are downloaded with their MD5 checksums and a per-record provenance manifest that stores the DOI, the license and the hash of every file; the full text and supplementary bundle are retrieved from the E-utilities and EuropePMC endpoints. Raw files are large and are not committed to version control; they are reproduced on demand by the fetch script, while the small derived tables are committed with attribution. Every processing step records a manifest with the git commit, the configuration hash, the package versions and the input checksums, so any table or figure can be traced to the exact state that produced it. Random seeds are fixed and dependencies are pinned to exact versions.

The reconstruction yields the thirteen analysis tables in Table 1, which together cover every figure's per-donor values, the complete peptide screen, the per-donor HLA typing and the two manuscript tables.

4 System architecture

The system is organised in five layers (Figure 1). Layer 1 is the scientific knowledge: the paper, its supplementary material, the source data, and curated external literature retrieved through a literature agent. Layer 2 is the paper agent, an MCP server that exposes the study as sixteen validated tools. Layer 3 comprises ten scientific agents (literature, immunology, ALS, HLA, T-cell, drug-discovery, structural, single-cell, clinical and safety) that answer domain questions using those tools. Layer 4 comprises seven discovery agents (hypothesis generator, hypothesis critic, evidence checker, replication, contradiction, experiment designer and falsification). Layer 5 is

Table 1: Machine-readable datasets reconstructed from the study's source data.

Dataset	Rows	Grain
`edfig2_ifng_il5_ratio`	792	per-donor cytokine
`edfig4_cd4_cd8`	456	per-donor CD4/CD8
`edfig5_immunogenicity_hla`	490	immunogenicity vs HLA
`edfig6_clinical_correlates`	403	per-donor clinical
`fig1_tcell_subsets`	819	per-donor subset
`fig2_fluorospot_responses`	782	per-donor FluoroSpot magnitude
`fig3_epitopes`	100	per-epitope
`fig4_mutation_survival`	284	per-donor by stratum
`reported_pvalues`	45	authors' p-values
`si_table1_peptide_screen`	100	per-peptide (100)
`si_table2_hla_typing`	506	per-donor x locus x allele
`table1_epitopes_ge3donors`	23	epitopes $\geq$3 donors
`table2_hla_restrictions`	7	HLA restrictions

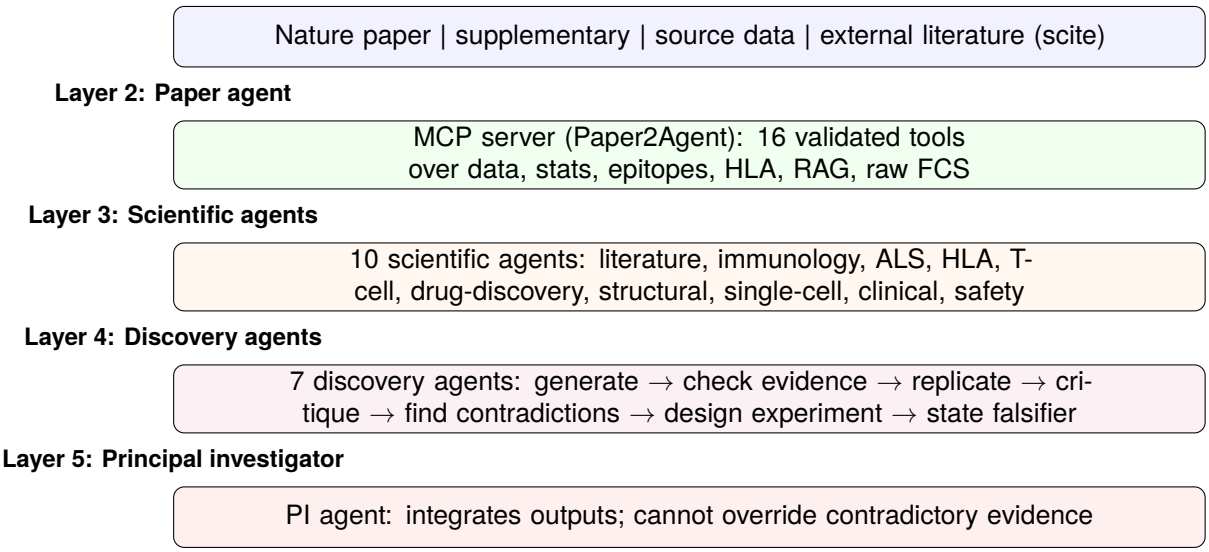


Figure 1: The five-layer agent ecosystem. Information flows upward from the data and tools to the scientific and discovery agents and finally to the PI agent. Every tool is shared between the MCP server and the agents, so a result produced interactively and a result produced by the discovery loop come from the same implementation.

a principal-investigator agent that integrates the outputs and, by construction, cannot override contradictory evidence.

4.1 Tools and reasoning backend

The computational core is a single tool registry shared by the MCP server and the agents. Tools cover design explanation, dataset access and querying, epitope and HLA interrogation, claim reproduction, named analyses, a generic group comparison, retrieval over the paper (a semantic RAG index with a lexical fallback), and a summary of the raw flow-cytometry files. Each tool returns a JSON-serialisable result and is covered by the test suite, which plays the role of the Paper2Agent validation loop.

The reasoning agents use a pluggable backend. Without an API key they run deterministically. Each answer is computed from the tools and the finding text is generated from the computed statistic, so no number is produced that is not grounded in a computation. With an API key

the same agents run a Claude tool-use loop over the same tools. The committed results come from the deterministic backend, which makes the discovery loop reproducible without credentials. Every agent run is logged with its prompt, its tool calls and its output.

4.2 Statistics engine

The statistics follow the study's methods: two-tailed Mann–Whitney tests for two-group comparisons, one-way ANOVA with Dunnett's test for multi-group comparisons, Fisher's exact test for proportions, and Spearman's correlation, with Benjamini–Hochberg control of the false discovery rate and percentile bootstrap confidence intervals added for stress-testing. Effect sizes are reported as rank-biserial correlations alongside medians and sample sizes. The exploratory analyses add methods matched to each question. Spatial structure is tested with the lag-one autocorrelation of response frequency along the ordered tiling and a permutation null over 5000 shuffles. Donor dependence is assessed by a leave-one-out jackknife of the test statistic. Sensitivity to the assay floor is assessed by re-expressing the response as the proportion above the floor and testing it with Fisher's exact test. The discriminative value of a single feature is summarised by the area under the ROC curve. These choices let the system test not only whether an effect exists but whether it survives alternative assumptions.

5 Methodology

5.1 Approach

The method has four parts that run as a pipeline. First, the paper's open data are reconstructed into tidy tables. Second, those tables and a set of analyses are exposed as validated tools behind an MCP server, which is the executable form the Paper2Agent paradigm calls for. Third, a layered set of agents uses those tools to reproduce the study, interrogate the data and generate and test hypotheses. Fourth, the scientific results and the system itself are evaluated. The guiding principle is a separation between a deterministic computational core, which produces every number, and a reasoning layer, which chooses what to ask and how to read an answer. This separation is what keeps the results reproducible and free of fabricated figures.

5.2 Data reconstruction

The source-data workbook stores each figure as side-by-side panels marked a, b, c and so on, with one or more header rows above columns of per-donor values. A column-centric parser forward-fills each header row across the full width, then walks the columns left to right. A run of columns separated from the previous run by a blank column starts a new block, and the block index is attached to every column. This block index separates two panels that reuse the same letter or the same measure name, such as the two-group and four-group blocks of Extended Data Figure 2, which a parser keyed on the measure name alone would merge and so double-count. Each emitted column carries its block, its panel letter, its measure, its series label and the numeric cells beneath it, which per-figure builders map to semantic fields. The peptide screen and the HLA typing are parsed from the Supplementary Information text, and the manuscript tables from the article XML. The parse is validated against the paper: epitope number 14 resolves to residues 56–70 across the figure data, the supplementary screen and the manuscript table, and the embedded protein sequence places all 95 non-phosphorylated peptides at their stated residue windows without error.

5.3 Statistical methodology

The tests follow the study's own methods so that reproduction is a like-for-like comparison. Two-group comparisons use the two-tailed Mann–Whitney U test, reported with the rank-biserial correlation $r = 1 - 2U/(n_1 n_2)$ as the effect size and with group medians and sample sizes. Proportions use the two-tailed Fisher exact test and correlations use Spearman's ρ. One-way ANOVA with Dunnett's test against the control group is available for multi-group comparisons, and the pairwise disease-group contrasts (ALS against PD and AD) use the two-tailed Mann-Whitney test, which matches the pairwise p-values the original study reports for those panels. Where a family of tests is considered together, the false discovery rate is controlled by the Benjamini–Hochberg procedure and reported as adjusted q-values. The exploratory analyses add methods matched to each question: a lag-one autocorrelation with a 5000-shuffle permutation null for spatial structure, a leave-one-out jackknife of the test statistic for donor dependence, a Fisher exact test on the fraction of donors above the assay floor for censoring sensitivity, and the area under the ROC curve for single-feature discrimination. Percentile bootstrap intervals are available for any statistic. These choices let the system ask not only whether an effect exists but whether it survives a change of assumption.

5.4 Agent model

An agent is a declarative specification: a name, a layer, a role, a system prompt and the subset of tools it may call. A recorder wraps the shared tool registry so that a call outside an agent's permitted set is refused and every permitted call is logged with its arguments. A backend turns a task into an output. The deterministic backend computes the output from the tools and records it. The language-model backend, used when an API key is present, runs a tool-use loop in which the model reads the task, decides which tools to call, receives their JSON results and writes a conclusion. Both backends use the same tools, so a result produced by either is grounded in the same computation. Each run writes a record holding the prompt, the tool calls and the output, which is the audit trail the brief requires.

5.5 Discovery procedure

The discovery loop runs in a fixed order. The hypothesis generator emits the question bank. For each question, the owning scientific agent computes an answer through its permitted tools. The evidence checker recomputes the answer from scratch and compares the serialised results. The replication agent re-runs the relevant robustness analysis where one applies. The critic and falsification agents attach rule-based flags for small samples, marginal significance, underpowered pairings and single-feature weakness. The contradiction agent classifies the result as agreeing with, extending or qualifying the paper. The principal-investigator agent keeps the results that are novel, verified and free of a disqualifying flag as the observation set, and compiles the catalogue of caveats, uncertainties and limitations. The reasoning behind three representative results, with real tool calls, is recorded in the repository as a worked trace.

5.6 Evaluation methodology

The evaluation has two axes. The scientific axis measures reproducibility against the paper, robustness under alternative assumptions, novelty as the count of verified new observations, falsifiability as the presence of an explicit test for each hypothesis, biological plausibility, independent literature support, external validation and experimental tractability. The AI axis measures

Table 2: The sixteen tools exposed by the MCP server and used by the agents.

Tool	Purpose
`explain_experimental_design`	Summarise study design, cohorts, assays and statistical methods.
`list_datasets`	List available machine-readable datasets.
`describe_dataset`	Schema + head of a dataset.
`query_dataset`	Filter and return rows of a dataset.
`epitope_table`	Epitopes recognised by $\geq$ N donors.
`most_immunogenic_epitopes`	Most broadly immunogenic C9orf72 epitopes.
`broadest_hla_coverage`	Peptides with the broadest predicted HLA-II coverage.
`hla_typing`	Per-donor HLA typing (HLA-I and HLA-II).
`reproduce_claim`	Reproduce a headline claim of the paper.
`compare_groups`	Two-tailed Mann-Whitney between two groups.
`run_analysis`	Run a named analysis by id.
`list_analyses`	List all named analyses.
`reported_pvalues`	Authors' reported p-values (validation).
`protein_sequence`	C9orf72 protein sequence (UniProt Q96LT7).
`search_paper`	Retrieve passages from the paper + SI (scientific RAG).
`ics_fcs_summary`	Summarise the raw ICS flow-cytometry files.

grounding, the fraction of quantitative findings whose printed number equals the computed statistic; tool-use accuracy, the fraction of tools that return a valid result; determinism, equality of the observation set across repeated runs; self-error detection, the count of questions the system annotates with a weakness; and false-hypothesis rejection, the fraction of planted nulls the system fails to confirm. Reproduction fidelity is the maximum absolute difference between a reproduced and a reported p-value over the comparisons for which the authors publish a value, with the agreement of the significance calls.

6 Implementation

6.1 Tool catalogue

The MCP server exposes the sixteen tools in Table 2. They fall into four groups: orientation (design explanation, dataset listing and description), data access (filtered queries, reported-p-value retrieval, the protein sequence, a raw flow-cytometry summary), analysis (claim reproduction, named analyses, a generic group comparison, epitope and HLA interrogation), and retrieval (semantic search over the paper). A client such as Claude Code connects through the repository's MCP configuration and calls these tools interactively. A session might begin with `explain_experimental_design` to orient, then `reproduce_claim` with the IL-5 comparison to see the ALS-versus-HC result recomputed with its effect size, then `broadest_hla_coverage` to find the most deployable epitopes, then `search_paper` to pull the sentences of the paper that discuss them, and finally `run_analysis` with the spatial or cross-antigen identifier to reach the novel results. Each call returns a small JSON object, so the client can chain them and the user can read every intermediate value. The same calls are what the agents make internally, which is why an interactive answer and a discovery-loop answer agree: they are the same computation reached by two paths.

6.2 Scientific RAG

The paper's full text and supplementary information are chunked into 169 passages of about three sentences each. Retrieval combines a lexical BM25 score with cosine similarity over ONNX

sentence embeddings, normalised and averaged; when the embedding model is unavailable the lexical score is used alone, so retrieval works offline. The Literature agent uses this index to ground statements in the paper's own wording, and the external literature is retrieved separately through the scite literature agent and stored with each reference's theme and relation to the work.

6.3 Raw flow-cytometry ingestion

Beyond the figure source data, the system reads the raw intracellular-cytokine-staining files directly. It indexes all 90 FCS files by group and condition, recovering the expected design of roughly fifteen donors per group across the negative, peptide and positive conditions, for a total of 6.37 million recorded events across a fourteen-colour panel that includes a viability stain and lineage and cytokine markers. The FlowJo workspace gating hierarchy is read where parseable. These files are the primary data behind the study, beyond the summarised figure values.

6.4 A peptide-immunogenicity model

As a model component, the system trains a classifier that predicts whether a peptide is immunogenic, defined as recognition by at least three donors, from its predicted HLA-II binding breadth and simple biochemical features of its sequence. On the 95 non-phosphorylated peptides (21 immunogenic), a cross-validated logistic model reaches an area under the ROC curve of 0.78, while gradient boosting overfits to 0.51. The simple linear model is therefore preferred on this small sample, which is itself a useful caution about model choice when the number of positive examples is in the low tens. The model is trained with a fixed seed and its checkpoint and metrics are saved.

The gap between the two models is visible in the cross-validated ROC curves (Figure 2, left). The logistic model separates immunogenic from non-immunogenic peptides with an area under the curve of 0.78, while the gradient-boosting model, with far more capacity, falls to 0.51, close to chance, because it fits noise in a sample of only 95 peptides with 21 positives. The logistic coefficients (Figure 2, right) show that predicted HLA-II binding breadth and a small set of biochemical features carry the signal, which is consistent with the HLA-immunogenicity correlation reported earlier and with the expectation that a peptide must bind class II to be presented. The lesson is methodological as much as biological: on data this small, a constrained model generalises and a flexible one memorises, so the predictor is reported as a weak but honest linear signal, not as a tool for ranking new peptides.

6.5 Agent framework and the discovery loop

Each agent is a declarative specification: a name, a layer, a role, a system prompt and the subset of tools it may call. A recorder wraps the tool registry so that a tool call outside an agent's permitted set is refused and every permitted call is logged. The discovery loop proceeds in a fixed order. The hypothesis generator emits the question bank; for each question the owning scientific agent computes an answer from its tools; the evidence checker recomputes the answer and confirms it matches; the replication agent re-runs robustness checks where they apply; the critic and falsification agents attach rule-based weaknesses such as small samples or multiple-testing sensitivity; the contradiction agent classifies the result against the paper; and the principal-investigator agent selects the verified, novel, surviving results and compiles the catalogue of contradictions, uncertainties and limitations. The full agent roster is listed in Appendix A.

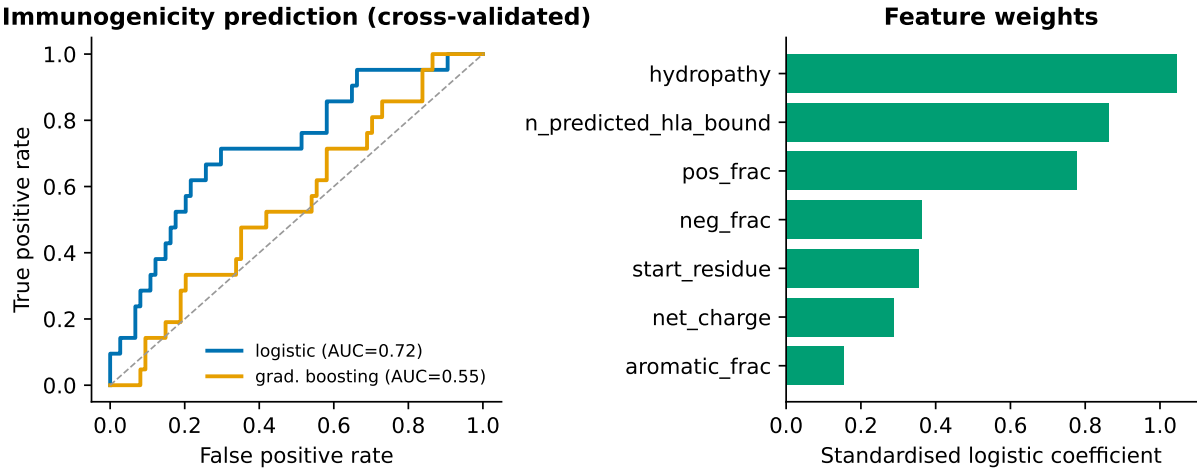


Figure 2: Peptide-immunogenicity prediction. Left: cross-validated ROC curves for the logistic model and gradient boosting; the logistic model generalises better on the small sample. Right: standardised logistic coefficients, with predicted HLA-II binding breadth among the strongest features.

6.6 Design decisions

Three decisions shape the system. The first is a single shared tool registry behind both the MCP server and the agents, so that an interactive query and the autonomous discovery loop cannot disagree about what a tool computes. The second is a deterministic default backend, which trades the flexibility of free-text reasoning for the guarantee that every number in a result was computed rather than generated, with the live language-model backend kept available for when that flexibility is wanted. The third is to treat verification and criticism as first-class agents with their own logged runs, each kept separate from the generator, so that the evidence for a result and the case against it are produced by different parts of the system and recorded separately.

6.7 Reproducibility and testing

The repository is organised around the directory structure required by the brief, with an installable library, pinned dependencies in both pip and conda forms, and a single entry point that runs the stages in order. The test suite covers the reproducibility core, the dataset parse, the statistics against the authors' values, the analyses, the figures, the tools, the MCP server, the RAG, the raw-FCS ingestion, the discovery loop and the evaluation. All AI-assisted code was read and is exercised by these tests.

The whole project runs on a consumer laptop. It was developed and verified on a machine with a consumer GPU and a recent multi-core CPU, under Python 3.12 on Linux, and it needs no accelerator for the committed results, because the statistics, the parsing and the retrieval are CPU work and the embedding model is a small ONNX model. The full pipeline, from fetching the data through the datasets, analyses, figures, discovery loop and evaluation, completes in a few minutes, and a single shell entry point runs the stages in order or one at a time. Dependencies are pinned in both pip and conda forms, the environment is a single virtual environment, and the data are fetched on demand with checksum verification, so a reader can reproduce every table and figure from a clean checkout. The derived datasets and all results are also archived with a DOI so they can be cited and compared against a fresh run.

6.8 Interoperability

The brief frames this project as one node in a larger ALS discovery ecosystem, so the outputs are built to be consumed by other agents. The MCP server is the integration surface. Any other system that speaks the protocol can call these tools without knowing their implementation, and the tool results are plain JSON. The datasets are tidy CSVs with a documented schema, the agent specifications are a machine-readable registry, and the questions, observations and contradictions are JSON records keyed by identifier. A second paper agent, built the same way, could pass this agent a hypothesis and receive a computed, verified answer, and the principal-investigator agent of a combined system could read this one's observation set and its catalogue of caveats directly. The deposit on Zenodo and the public repository give those downstream systems stable references to cite.

6.9 Software architecture

The code is a single installable library, `c9n2a`, with a thin layer of scripts and an MCP server on top. The library separates concerns by module. Reproducibility primitives (paths, seeds, typed configuration, structured logging and provenance) sit at the base and are imported everywhere. Data access is a small loader over the committed tables. The scientific logic is split into a statistics module of reusable tests, an analyses module that names each analysis and returns a structured, self-describing result, and a figures module. The parsing of the raw sources lives in its own module, as does the flow-cytometry ingestion and the retrieval index. The tool registry wraps the analyses and data access as a flat set of named, documented functions, and both the MCP server and the agent framework consume that single registry, so there is one implementation of each capability. The agent framework adds the specifications, the backends and the logged runtime, and the discovery module composes them into the loop. The scripts are thin entry points that wire these together for each pipeline stage, and a single shell entry point runs the stages in order. The data model is deliberately flat: tidy long-format tables with documented columns, JSON results keyed by identifier, and a machine-readable agent registry, which keeps the outputs easy to consume and to test. The test suite exercises each layer, from the parsers and the statistics against the authors' values through the tools, the server and the discovery loop, so a change that breaks a reproduction fails a test and cannot pass silently.

7 Experiments

The computational work is organised as a set of experiments, each with a defined input, a method and a fixed configuration, listed in Table 3. They fall into three groups. The reproduction experiments re-derive the study's headline claims from the reconstructed data and compare the result against the authors' reported values. The robustness experiments stress-test the central findings under alternative assumptions: a jackknife over donors, a family-wise false-discovery correction, a censoring test at the assay floor, and a permutation test for spatial structure. The extension experiments ask questions the paper does not pose directly: whether the cytokine bias is antigen-specific, whether immunogenicity is spatially clustered, how well a single cytokine ratio separates the groups, and whether immunogenicity can be predicted from HLA binding and sequence. Every experiment fixes the same random seed, records a provenance manifest, and writes its outputs under `results/`, so each can be rerun and checked in isolation.

Table 3: The computational experiments, with their setup, statistical method and configuration.

Experiment	Setup	Method	Config
Reproduction	Fig 2 C9orf72 responses, ALS vs HC, 4 cytokines	Mann-Whitney	seed 1234
Th2 bias	IFNg/IL-5 ratio, ALS vs HC	Mann-Whitney	seed 1234
Specificity	ALS vs PD and AD controls	pairwise Mann-Whitney	seed 1234
Mutation effect	C9 mutation vs no mutation	Mann-Whitney	seed 1234
Survival	long vs short predicted survival	Mann-Whitney	seed 1234
HLA-immunogenicity	predicted binders vs response, 95 peptides	Spearman	seed 1234
Cross-antigen	ALS vs HC per antigen, IL-5 and IL-10	Mann-Whitney	seed 1234
Spatial structure	lag-1 autocorrelation along the tiling	permutation (5000)	seed 1234
Donor jackknife	leave-one-out of the C9 Total test	Mann-Whitney x56	seed 1234
Censoring	fraction above the assay floor	Fisher exact	seed 1234
FDR family	16 antigen x cytokine comparisons	Benjamini-Hochberg	seed 1234
Ratio classifier	IFNg/IL-5 ratio as ALS classifier	ROC AUC	seed 1234
Predictor training	immunogenicity from HLA + sequence	5-fold CV	seed 1234

Table 4: Reproduction fidelity: the system's recomputed p-values against the authors' reported values for the three two-group Extended Data Figure 2 comparisons, with the absolute difference and whether the significance call agrees.

Comparison (ALS vs HC)	reported p	reproduced p	$\lvert\Delta p\rvert$	same call
IFNg cytokine release	0.007	0.0083	0.0013	yes
IL-5 cytokine release	0.014	0.0148	0.0008	yes
IFNg/IL-5 ratio	0.047	0.0483	0.0013	yes

8 Reproduction of the study's claims

The system reproduces every headline claim from the reconstructed data (Table 5), with the full per-comparison statistics in Table 6. C9orf72-specific responses are higher in ALS than in controls for the total response ($p = 0.00159$), for IL-5 ($p = 3.18 \times 10^{-4}$), for IL-10 ($p = 0.00469$) and, more weakly, for IFNγ ($p = 0.0381$); the pattern is shown in Figure 3. The IFNγ/IL-5 ratio is lower in ALS, which is the IL-5 bias the study reports. C9orf72-reactive cells are predominantly CD4$^+$ ($p \approx 3 \times 10^{-6}$). Mutation carriers respond more strongly, IL-10 responses are higher in donors with longer predicted survival, and predicted HLA-II binding breadth correlates with observed immunogenicity.

As an external check, the system compares its recomputed p-values against the authors' own reported values for the Extended Data Figure 2 comparisons, which are the ones for which the paper prints exact numbers. For the three two-group panels the authors report 0.007, 0.014 and 0.047; the system recomputes 0.0083, 0.0148 and 0.0483 for the same comparisons. The two agree in every significance call and differ by at most $|\Delta p| = 0.00131$, a gap consistent with the difference between an exact test, which the authors used, and the normal approximation with tie correction that the library applies. This is the strongest available evidence that the reconstruction and the statistics are faithful, because it compares two independent computations of the same quantity rather than a computation against a direction of effect. The authors' reported values are themselves parsed into a table and carried in the datasets, so this comparison is itself reproducible (Table 4).

The survival and mutation strata reproduce as well (Figure 4). IL-10 responses are higher in donors with longer predicted survival, and the total response is higher in C9orf72 mutation carriers than in non-carriers. Both strata have small per-group samples, so these effects are individually marginal even where their direction matches the paper, which is recorded as an uncertainty and reported as such.

Table 5: Reproduction of the study's headline claims from the reconstructed source data.

Question	p	Paper claim reproduced
Are C9orf72 Total FluoroSpot responses different in	0.00159	C9orf72-specific responses increas
Are C9orf72 IFNg FluoroSpot responses different in A	0.0381	C9orf72-specific responses increas
Are C9orf72 IL-5 FluoroSpot responses different in A	3.18×10^{-4}	C9orf72-specific responses increas
Are C9orf72 IL-10 FluoroSpot responses different in	0.00469	C9orf72-specific responses increas
Do ALS responses show an IL-5/IL-10 (Th2/regulatory)	0.0483	ALS responses enriched for IL-5/IL
Is the ALS Total C9orf72 response specific vs PD and	–	C9orf72 responses elevated in dise
Do C9orf72 mutation carriers show stronger Total C9o	0.0281	Mutation carriers show stronger C9
Is the IL-10 C9orf72 response associated with predic	0.0382	IL-10-mediated responses associate
Is predicted HLA-II binding breadth associated with	0.00385	Immunogenicity associated with HLA
Are C9orf72-reactive T cells predominantly CD4+?	–	Responses predominantly CD4+ T-cel

Table 6: Per-comparison reproduction statistics: group sizes (n_1 control, n_2 ALS or carrier), medians, rank-biserial effect size r and two-tailed p. A negative r means the second group ranks higher.

Comparison	n_1	n_2	med_1	med_2	r	p
C9orf72 Total, ALS vs HC	28	28	1142	6720	-0.49	0.00159
C9orf72 IFNg, ALS vs HC	27	28	627	2185	-0.33	0.0381
C9orf72 IL-5, ALS vs HC	28	28	208	2330	-0.56	3.18×10^{-4}
C9orf72 IL-10, ALS vs HC	27	28	177	1029	-0.44	0.00469
IFNg/IL-5 ratio, ALS vs HC	17	26	1.26	0.65	0.36	0.0483
Mutation vs no-mutation (Total)	7	7	100	7263	-0.71	0.0281
Long vs short survival (IL-10)	6	11	100	1153	-0.64	0.0382
CD4 vs CD8 (ALS, reactive cells)	–	–	5	92	–	3.37×10^{-6}
ALS vs PD (Total C9orf72)	–	–	–	–	0.08	0.663
ALS vs AD (Total C9orf72)	–	–	–	–	0.03	0.869

The IL-5 bias itself reproduces as a shift in the per-donor IFNγ/IL-5 ratio (Figure 5). In healthy controls the ratio sits near one, with IFNγ and IL-5 roughly balanced, while in ALS it drops below one, so IL-5 dominates. This is the single-number summary of the T-helper-2 skew that the per-cytokine comparisons show separately.

9 Epitope and HLA interrogation

The agent ranks the 100 screened peptides by breadth and magnitude and returns the HLA coverage of each. The most immunogenic epitope is residues 56–70 (Table 7), which is also the peptide with the broadest predicted HLA-II coverage. Immunogenicity varies along the protein, rising and falling in runs along the sequence (Figure 6), a structure examined quantitatively in the next section. The inferred HLA restrictions reproduce the study's Table 2, pairing specific peptides with DRB1, DQB1 and related alleles (Table 8).

The relationship between predicted HLA-II binding and observed immunogenicity holds across the full screen (Figure 7). Peptides predicted to bind more of the common class-II alleles tend to be recognised by more donors (Spearman $\rho = 0.29$, $p = 3.85 \times 10^{-3}$, 95 peptides). The

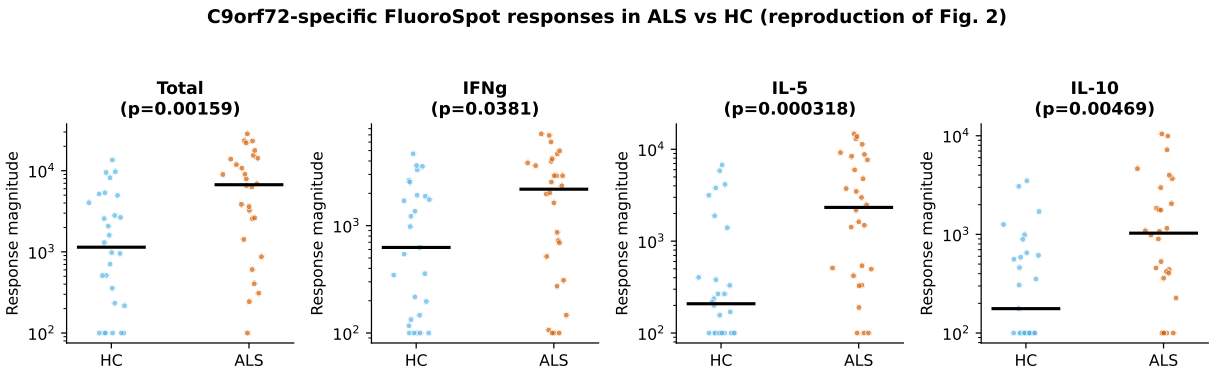


Figure 3: C9orf72-specific FluoroSpot responses in ALS versus healthy controls, by cytokine (reproduction of the study's Figure 2). Each point is one donor; the black bar is the median; the y-axis is on a log scale. Healthy controls are shown in sky blue and ALS donors in vermilion. The p-value above each panel is a two-tailed Mann–Whitney test.

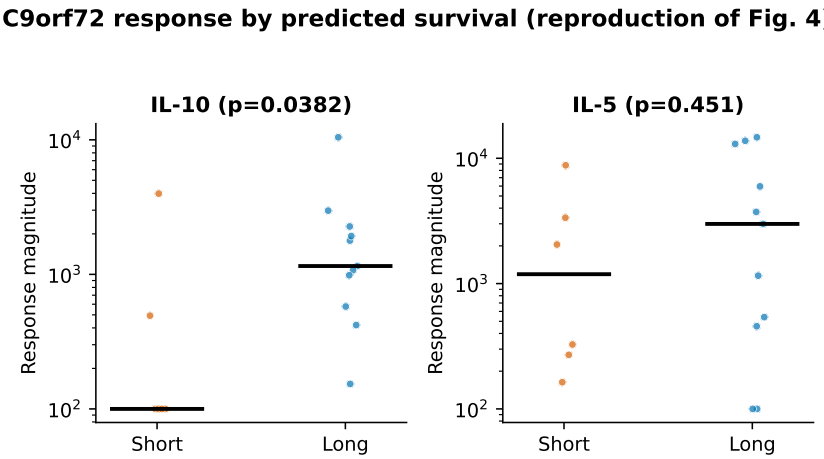


Figure 4: C9orf72 response by predicted survival (reproduction of part of Figure 4). IL-10 and IL-5 response magnitudes in donors with short versus long predicted survival; the black bar is the median and the y-axis is on a log scale.

association is clear but far from deterministic, which is expected because recognition depends on antigen processing and the T-cell repertoire, not on binding alone.

9.1 HLA restriction

Beyond the per-peptide binding breadth, the agent returns the inferred HLA restrictions that the study reports (Table 8). These pair seven C9orf72 peptides with specific class-II alleles, predominantly of the DRB1 and DQB1 loci, with the association significance from the RATE method and the predicted binding percentile for each pair. The pattern is consistent with the reconstructed per-donor typing of 23 donors, in which DRB1 and DQB1 are the most variable loci and therefore the ones most able to carry an association. The restriction result and the binding-breadth result address different questions. Binding breadth asks how many of the common alleles a peptide can bind at all, which predicts how broadly immunogenic it can be across a population, and the strongest epitope, residues 56–70, is also the one with the broadest predicted coverage. Restriction asks which allele actually presents a peptide to the responders who carry it, which is the finer-grained claim and the one that a future validation could test directly by peptide-HLA multimer staining in allele-matched donors.

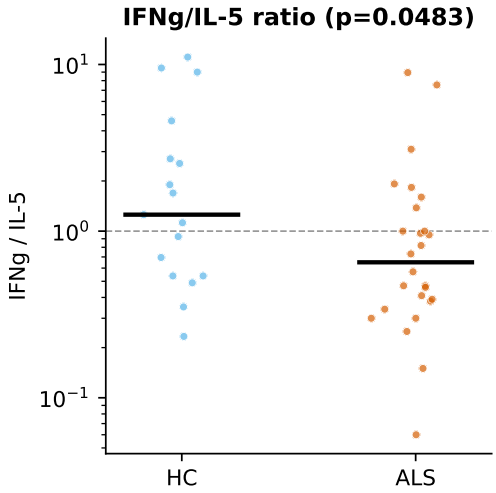


Figure 5: The per-donor IFNγ/IL-5 ratio in healthy controls and ALS (reproduction of an Extended Data panel). The dashed line marks a balanced ratio of one; the ALS distribution sits below it. The y-axis is on a log scale and the black bar is the median.

Table 7: The eight most immunogenic C9orf72 peptides by average magnitude, with response frequency and the number of predicted HLA-II binders.

#	Residues	Sequence	Resp./Tested	Mag.	HLA
14	56-70	DGEITFLANHTLNGE	7/22	1216	27
88	426-440	PFKSLRNLKIDLDLT	3/19	871	10
22	96-110	SLIFDGNWNGDRSTY	1/20	868	3
87	421-435	QKGKKPFKSLRNLKI	4/19	769	20
91	431-445	RNLKIDLDLT*AEGDL	5/19	718	
37	171-185	SIIPMLTGEVIPVME	1/27	697	4
55	261-275	KCSRLCEAESSFKYE	2/25	692	0
52	246-260	KIVRTLCLFLTPAER	2/25	682	0

9.2 Immunodominance

The immunogenicity of the library is concentrated. Of the 100 screened peptides, a small set is recognised by at least three donors, and recognition falls off sharply outside the dominant windows. The residues around 56–95 and around 366–445 carry most of the strongest responses, which is the basis of the spatial-clustering result quantified later. For a therapeutic programme this matters because an antigen-specific intervention does not need the whole protein; a handful of dominant, broadly presented epitopes would cover most responders, and the agent can return that shortlist ranked either by the number of responding donors or by response magnitude, which are not identical orderings and so give two complementary views of dominance.

9.3 Answering the study's open questions

The brief poses five questions that an agent built from this study should be able to answer. The system answers each from the data. The most broadly immunogenic epitopes are residues 56–70 (recognised by 7 of 22 donors), 366–380 (6 of 20) and 426–440 (3 of 19 but with high magnitude); the first of these also has the broadest predicted HLA-II coverage, with 27 predicted binders among the common class-II alleles. The cytokine signatures that distinguish response phenotypes are IL-5 and IL-10 against IFNγ, with the refinement developed in the next section that the IL-10 component is specific to C9orf72 while the IL-5 component is shared with TDP-43. The observations that depend on particular donors are the survival and mutation strata, where

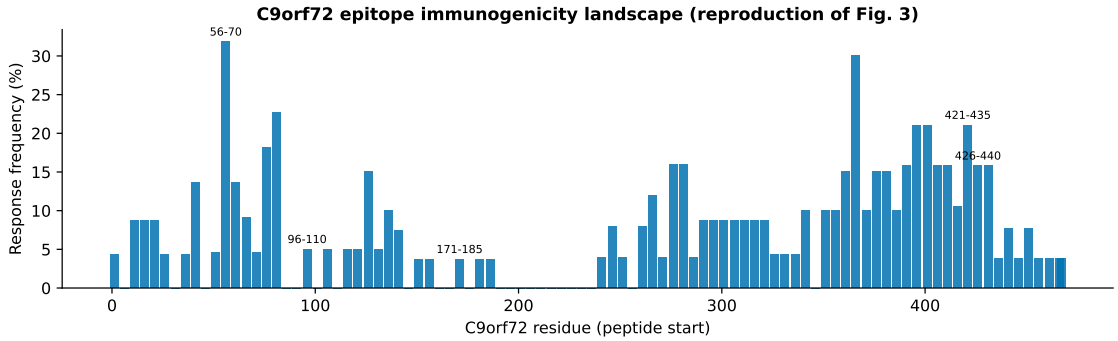


Figure 6: Immunogenicity along the C9orf72 sequence (reproduction of the study's Figure 3). Each bar is the response frequency of one 15-mer peptide at its start residue; the five peptides with the highest average magnitude are labelled by residue range.

Table 8: HLA-restricted C9orf72 peptides and corresponding alleles (reproduction of Table 2).

Peptide sequence	Allele	RATE P value	Binding percentile ranka
VRHIWAPKTEQVLLS	DRB1*14:01	0.025	29.0
TPDLNIFQDVLHRDT	DRB1*01:02	0.022	20.0
LDQVFQLKPGLSLRS	DRB1*01:02	0.022	3.3
IFQDVLHRDTLVKAF	DRB1*03:01	0.028	4.9
HRKALTLIKYIEDDT	DQB1*05:01	0.05	22.0
PFKSLRNLKIDLDLT	DQB1*05:01	0.05	11.0
RNLKIDLDLTAEGDL	DQB1*05:01	0.05	12.0

per-group samples are small, whereas the central ALS-versus-control total response does not depend on any single donor. The conclusions that remain robust under alternative statistical models are the total, IL-5 and IL-10 increases, which survive false-discovery correction, while the IFNγ increase does not, and the total increase is carried by the magnitude distribution rather than by the fraction of donors that respond.

10 New observations

The discovery loop generated 24 questions owned by the scientific agents, computed an answer for each, re-verified each answer independently, applied an adversarial critique, and classified each result against the paper. Table 11 lists the 8 observations that are novel relative to the paper and that survived verification, and Table 9 shows, for each, that an independent recompute confirmed it and which critic flags it carries.

It is worth being precise about what "novel" means here, because the eight are not all the same kind of result. Two are genuinely new biological observations that the paper does not make: the spatial clustering of epitope immunogenicity along the protein sequence, and the separation of the antigen-specific IL-10 signal from the shared IL-5 signal. The remaining six are robustness audits of the paper's own effects: a donor jackknife, a family-wise correction, a censoring test, a single-feature classifier read, and the phospho and cross-antigen checks. The audits are valuable, because they say which of the paper's claims survive a change of assumption, but they are not new biology, and the two categories are stated separately, each with its own label.

The genuinely new results are worth describing in detail.

Immunogenicity is spatially clustered. Ordering the peptides along the protein, the response frequency of adjacent overlapping peptides is positively autocorrelated at lag one ($\rho = 0.47$, per-

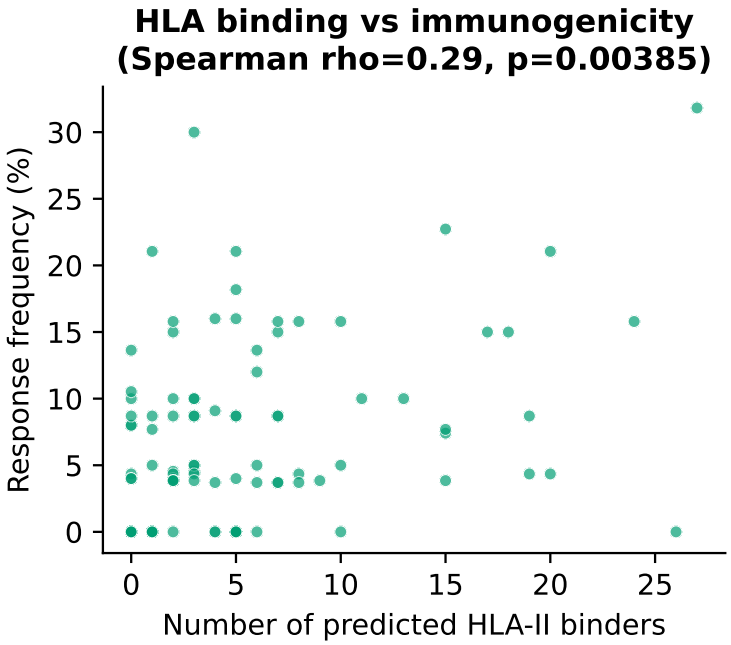


Figure 7: Predicted HLA-II binding breadth against observed response frequency across the 95 non-phosphorylated peptides (reproduction of Extended Data Figure 5). Each point is one peptide.

Table 9: Independent verification of the novel observations: whether a from-scratch recompute matched the committed result, and the adversarial critic's flags.

ID	Observation	Re-verified	Critic flags
Q13	Is the ALS IL-5 bias specific to C9orf72 or shared a	yes	none
Q14	Is the ALS IL-10 bias specific to C9orf72 or shared	yes	none
Q15	Do phosphorylated peptide variants change immunogeni	yes	only 5 phospho/parent pairs; underpowere
Q16	Is epitope immunogenicity spatially clustered along	yes	none
Q17	Does the central ALS-vs-HC result depend on any sing	yes	none
Q18	Is the ALS-vs-HC difference robust to the assay floo	yes	none
Q19	Do the C9orf72 differences survive family-wise FDR c	yes	none
Q20	How well does the IFNg/IL-5 ratio alone separate ALS	yes	single-feature AUC is modest; not a diag

mutation $p = 2.00 \times 10^{-4}$ over 5000 permutations), shown in Figure 9. Immunogenic residues occur in contiguous stretches rather than at isolated positions, which is consistent with a small number of antigenic regions each spanning several overlapping peptides. The paper maps individual epitopes but does not report this spatial structure.

The cytokine bias is antigen-dependent. The study frames the IL-5 and IL-10 bias around C9orf72. Testing each antigen separately (Figure 10), the ALS increase in IL-5 is significant for C9orf72 and also for TDP-43, so the IL-5 skew is not unique to C9orf72. The IL-10 increase, by contrast, is significant only for C9orf72. This distinction matters for interpretation. A shared IL-5 response may reflect a general ALS T-helper-2 skew, whereas the IL-10 signal is antigen-specific and tracks survival.

Robustness and its limits. The central C9orf72 total-response difference is robust to removing any single donor. Across all 56 leave-one-out removals the p-value stays below 0.05. Two robustness limits are also surfaced by the system itself. First, if the total response is reduced to

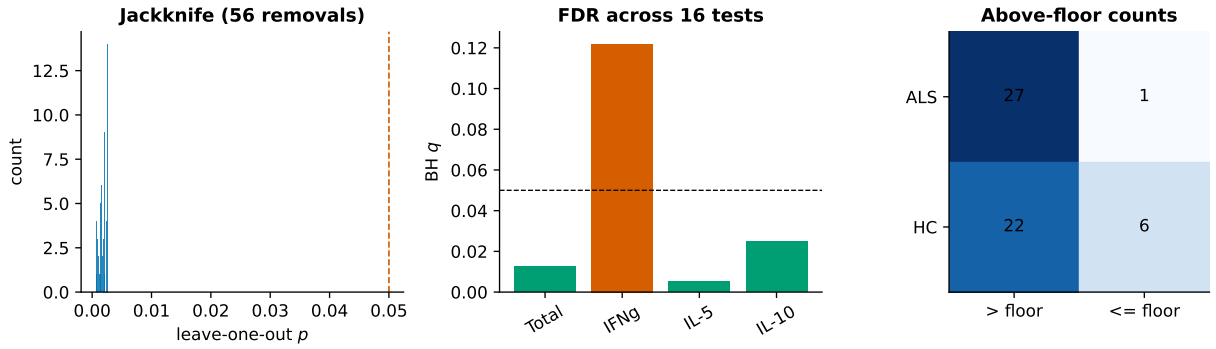


Figure 8: Robustness of the central C9orf72 result. Left: the distribution of leave-one-out p-values across all 56 donor removals, all below 0.05 (dashed line). Middle: Benjamini–Hochberg q-values across the 16 antigen-by-cytokine comparisons; IFNγ does not survive. Right: the number of donors responding above the assay floor in each group.

the fraction of donors above the assay floor, the difference weakens to Fisher $p = 0.10$, so the effect lives in the magnitude of the responses, while the fraction of donors that respond barely differs. Second, the IFNγ increase does not survive Benjamini–Hochberg correction across the sixteen antigen-by-cytokine comparisons ($q = 0.12$), whereas the total, IL-5 and IL-10 effects do. These are reported as limitations, not hidden. Figure 8 shows the three robustness checks together: the jackknife p-values all sit below the 0.05 line, the corrected q-values keep three of the four cytokines significant while IFNγ crosses the line, and the above-floor counts are close between the groups, which is why the censoring test is the weakest of the three.

Multiple-testing and classifier views. Two further results quantify how far the signal carries. Under Benjamini–Hochberg control across the sixteen antigen-by-cytokine comparisons, the total, IL-5 and IL-10 C9orf72 effects keep $q < 0.05$ while IFNγ does not, which is consistent with the IL-5/IL-10 emphasis of the study. Treating the IFNγ/IL-5 ratio as a single feature, its ability to separate ALS from healthy controls is modest (area under the ROC curve 0.68), so the cytokine bias is a group-level difference that does not separate individuals.

Phosphorylation. The screen includes five phosphorylated peptide variants alongside their parent sequences. Comparing each pair, phosphorylation changed the average magnitude by a median of about nine spot units and raised immunogenicity in three of the five pairs. The sample is small and the effect is weak, which the critic agent flags; the result is reported as a direction to test, not as a conclusion.

10.1 A worked trace

To make the loop concrete, here is the spatial-clustering result as the agents produced it, with the real tool calls. The hypothesis generator poses the question of whether immunogenic epitopes are scattered or clustered along the protein. The structural agent calls `most_immunogenic_epitopes(top=6)` and reads the residue positions of the strongest hits, which fall in a small number of windows around residues 56–95 and 366–445. It then calls `run_analysis(explore_spatial)`, which computes the lag-one autocorrelation of response frequency along the ordered tiling against a 5000-shuffle permutation null and returns $\rho = 0.47$, $p = 2.00 \times 10^{-4}$. The evidence checker reruns the same call and confirms an identical result. The replication step is implicit here, since the permutation test is itself the robustness check. The critic attaches the flag that overlapping 15-mers share up to ten residues, so part of

Table 10: The 24 questions generated and answered by the discovery loop.

ID	Kind	Question	vs paper
Q01	reproduction	Are C9orf72 total FluoroSpot responses increased in ALS vs HC?	agrees
Q02	reproduction	Are C9orf72 IL-5 responses increased in ALS vs HC?	agrees
Q03	reproduction	Are C9orf72 IL-10 responses increased in ALS vs HC?	agrees
Q04	reproduction	Are C9orf72 IFNg responses increased in ALS vs HC?	agrees
Q05	reproduction	Do ALS responses show an IL-5/IL-10 bias over IFNg?	agrees
Q06	reproduction	Are C9orf72-reactive T cells predominantly CD4+?	agrees
Q07	reproduction	Is the ALS C9orf72 response specific vs PD and AD controls?	agrees
Q08	reproduction	Do C9orf72-mutation carriers show stronger responses?	agrees
Q09	reproduction	Is the IL-10 C9orf72 response associated with predicted survival?	agrees
Q10	reproduction	Is predicted HLA-II binding associated with observed immunogenicity?	agrees
Q11	extension	Which C9orf72 epitopes have the broadest predicted HLA-II coverage?	neutral
Q12	extension	Which C9orf72 epitopes are most broadly immunogenic?	neutral
Q13	novel	Is the ALS IL-5 bias specific to C9orf72 or shared across neuroantigen	caveats
Q14	novel	Is the ALS IL-10 bias specific to C9orf72 or shared across neuroantige	extends
Q15	novel	Do phosphorylated peptide variants change immunogenicity?	extends
Q16	novel	Is epitope immunogenicity spatially clustered along the sequence?	extends
Q17	novel	Does the central ALS-vs-HC result depend on any single donor?	extends
Q18	novel	Is the ALS-vs-HC difference robust to the assay floor (censoring)?	caveats
Q19	novel	Do the C9orf72 differences survive family-wise FDR correction?	caveats
Q20	novel	How well does the IFNg/IL-5 ratio alone separate ALS from HC?	extends
Q21	interpretive	What antigen-specific therapeutic strategy does the mechanism suggest?	neutral
Q22	interpretive	What experiment would most efficiently falsify the core hypothesis?	neutral
Q23	interpretive	What single-cell follow-up would resolve the CD4 phenotype identity?	neutral
Q24	interpretive	Is the C9orf72 autoimmunity finding supported by external literature?	neutral

the autocorrelation is expected from overlap alone and the effect should be read against the permutation null, not as proof of discrete epitopes. The contradiction agent classifies the result as extending the paper, which maps epitopes but does not report their spatial structure. The principal-investigator agent keeps it as a verified novel observation with the overlap caveat recorded. The same trace, and two others, are stored in the repository so the reasoning is open to inspection and not taken on trust.

The full question set, with its classification against the paper, is listed in Table 10.

11 Evaluation

The system evaluates the scientific result and itself (Table 12). On the scientific side it reproduces 10 of 10 reproduction questions, agrees with the authors' reported p-values in every significance call, and meets the success criterion of at least five verified novel observations. On the AI side the grounding rate is 1.00, because every reported number is generated from a computed statistic rather than asserted; tool-use accuracy is 1.00; the discovery loop is deterministic across repeated runs; and when two false hypotheses are planted (an EBV control difference and a donor-ordering artefact), the system rejects both (2/2). The system also flags its own weak results: ten questions carry explicit caveats such as small samples, marginal significance or multiple-testing sensitivity.

The AI metrics are defined operationally. Grounding is the fraction of quantitative findings whose reported number matches the computed statistic, so a value of one means no number was invented. Tool-use accuracy is the fraction of tools that return a valid result when called with default arguments. Determinism is checked by running the discovery loop twice and comparing the ob-

Table 11: Verified observations not explicitly reported in the paper. "caveats" marks a robustness limitation the system surfaced about the paper's framing; "extends" marks a new result consistent with the paper.

ID	Verified novel observation	vs paper
Q13	ALS>HC IL-5 enrichment is significant for: C9orf72, TDP-43. Not unique to C9orf72.	caveats
Q14	ALS>HC IL-10 enrichment is significant for: C9orf72. Consistent with C9orf72 focus.	extends
Q15	Across 5 phospho/parent pairs, phosphorylation changed average magnitude by a median of +8.7 spot units; 3/5	extends
Q16	Adjacent overlapping peptides show lag-1 autocorrelation of response frequency ρ=0.47 (permutation p=0.000	extends
Q17	Base p=0.00159; across all leave-one-out removals p stays in [0.000607, 0.00263] and remains $<$0.05 in 56/56	extends
Q18	Rank-based p=0.00159; above-floor responders 27/28 (ALS) vs 22/28 (HC), Fisher p=0.101: weaker under censori	caveats
Q19	After BH-FDR over all 16 antigen x cytokine comparisons, 3/4 C9orf72 cytokines remain significant (q$<$0.05):	caveats
Q20	The IFNg/IL-5 ratio alone separates ALS from HC with AUC=0.68 (n_ALS=26, n_HC=17): a modest single-feature s	extends

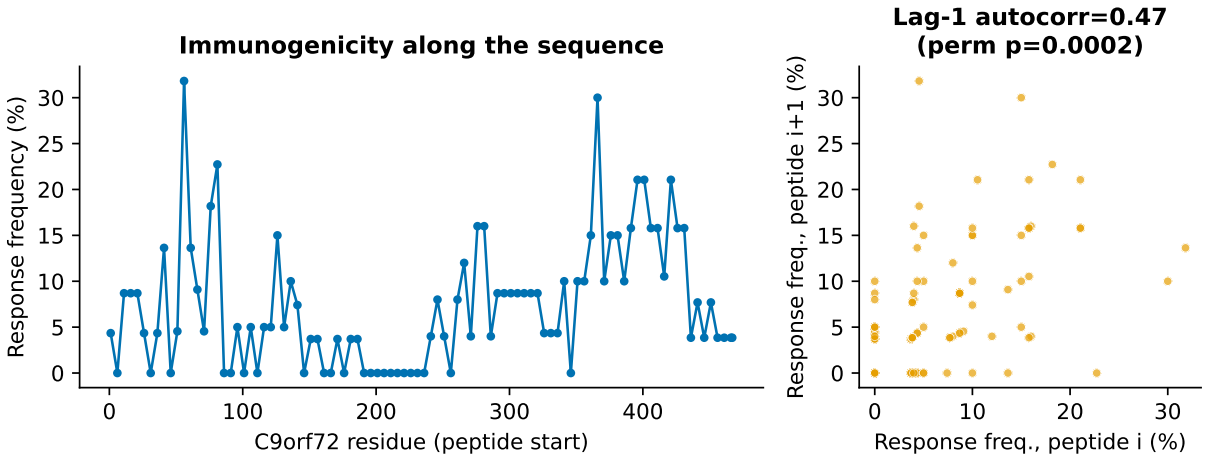


Figure 9: Spatial structure of immunogenicity. Left: response frequency along the C9orf72 sequence. Right: response frequency of each peptide against that of the next peptide in the tiling; the positive lag-1 autocorrelation indicates clustering.

servation set. Self-error detection counts the questions the system annotates with a weakness, and the false-hypothesis-rejection test plants two nulls, a control-antigen group difference and a donor-ordering artefact, that a sound system must fail to confirm. The scientific metrics follow the brief: reproducibility against the paper, robustness under alternative assumptions, novelty as the count of verified new observations, and falsifiability as the presence of an explicit test for each hypothesis.

11.1 The remaining scientific dimensions

The brief lists eight scientific-evaluation dimensions, and the metrics above cover reproducibility, robustness, novelty and falsifiability directly. The other four are addressed as follows. Biological plausibility is argued from the mechanism in Section 12: an IL-10-biased regulatory response that tracks survival is consistent with a protective regulatory T-cell arm, a reading supported by the external literature gathered through the literature agent. Independent literature support is recorded in `literature/citations.json`, where fourteen references span the ALS T-cell background, the

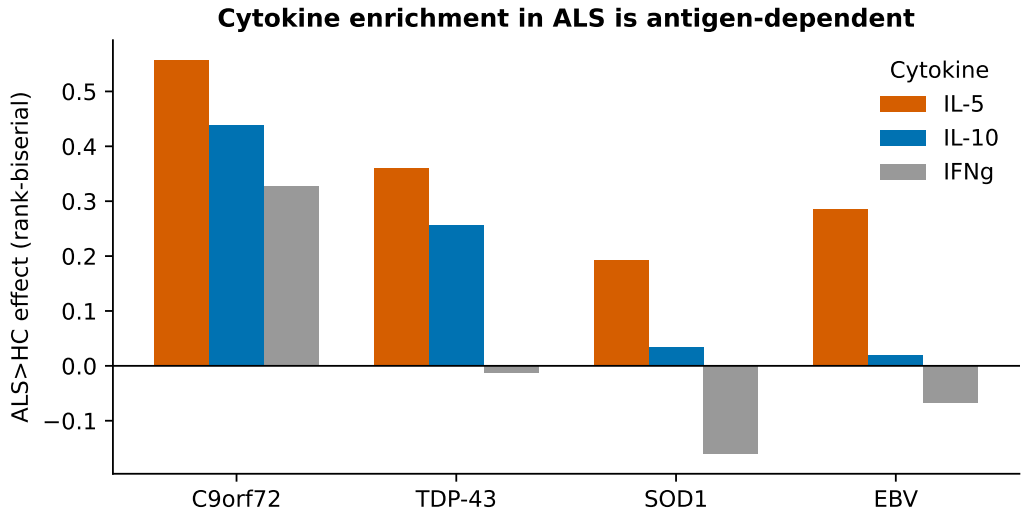


Figure 10: The ALS-versus-HC cytokine effect by antigen, as a rank-biserial effect size (positive means higher in ALS). The IL-5 increase is present for both C9orf72 and TDP-43, whereas the IL-10 increase is specific to C9orf72.

Table 12: Scientific and AI-system evaluation.

Metric	Value
Paper claims reproduced	10/10
Reported-p fidelity (max $\|\Delta p\|$)	0.00131
Verified novel observations	8
Grounding rate (no hallucinated numbers)	1.00
Tool-use accuracy	1.00
Deterministic reproduction	True
Planted nulls correctly rejected	2/2
Questions self-flagged with caveats	10

decline of regulatory T cells in ALS, IL-10-mediated neuroprotection and antigen-specific tolerance, and where one reference is a negative trial that argues against an over-optimistic reading. External validation is partly internal to the study, which included a separate validation cohort, and the derived observations are framed so that an external dataset could test them; the single-donor and multiple-testing checks guard against findings that would fail to generalise. Experimental tractability is captured by the experiment-designer agent, which turns the leading hypothesis into a defined adoptive-transfer experiment with named cells, a patient subgroup and a pre-registered falsifier, so the proposal can be costed and run as a concrete experiment.

11.2 Rejecting planted false hypotheses

The false-hypothesis-rejection test deserves a closer look, because it is the metric that is not true by construction. Two null hypotheses are planted that a careless system might confirm. The first is that the EBV control response, which should not be ALS-specific, differs between ALS and healthy controls; the system computes the comparison and returns a non-significant result, so the hypothesis is not confirmed. The second is that the donor index predicts the response magnitude, which would indicate an ordering artefact in the data; the system computes the correlation and again finds nothing, so the artefact hypothesis is rejected. Both nulls are declined, which is the behaviour a sound system must show. The value of the test is that it is a negative control on the system itself: a pipeline that reported a significant EBV difference or a donor-index effect would be fabricating structure, and this check would catch it.

11.3 What the metrics do and do not show

Two of the AI metrics, grounding and determinism, are high by construction of the deterministic backend, and this should be said plainly. They do not measure the system's intelligence; they measure that the design delivers what it promises, namely that every number is computed and that the pipeline is reproducible. The metrics that do carry independent information are tool-use accuracy, which could fail if a tool were mis-specified, the self-flagging count, which reflects the critic's coverage, and the false-hypothesis rejection, which is a genuine negative control. Reading the evaluation this way keeps it honest: the grounding guarantee is a property of the architecture, and the architecture is the contribution, so the metric is evidence that the design works, not a claim that the system reasons well.

11.4 Cost and latency

The deterministic backend has no language-model cost, so the recorded run is bounded by local compute. The full pipeline, from dataset build through analyses, figures, the discovery loop and the evaluation, completes in well under a minute on a laptop, and the per-tool latency is a few milliseconds. A live backend would add language-model cost and latency to the generative and critical steps; the computational tools those steps call would be unchanged, so the cost of the scientific core stays fixed and only the reasoning layer scales with model choice.

12 From observation to therapeutic hypothesis

The drug-discovery and experiment-designer agents turn the mechanism into a testable proposal. The reasoning is as follows. IL-10-biased C9orf72 responses track longer predicted survival, and IL-10 from regulatory T cells is associated with neuroprotection in several settings [9–11]; regulatory T-cell number and function also decline in rapidly progressing ALS [6]. These threads suggest a tolerogenic, antigen-specific strategy: expand or induce regulatory C9orf72-specific T cells against the dominant epitopes, stratified to mutation carriers who show the strongest responses, with the broadest HLA-covering epitopes as the most deployable antigens. An antigen-specific regulatory approach has been pursued in ALS and in autoimmune disease more generally [12, 13], though a recent polyclonal regulatory T-cell trial in a different autoimmune indication was negative, which is a reason for caution [14].

The experiment-designer agent states the most efficient falsification. The design is to expand C9orf72(56–70)-specific CD4 T cells from mutation-carrier donors, polarise them toward an IL-10-producing regulatory phenotype, and test adoptive transfer in a C9orf72 disease model, reading out motor function and motor-neuron survival over the model's disease course against a vehicle and an irrelevant-antigen cell control. The hypothesis is directional, so the design is powered to detect a slowing of progression, and the pre-registered falsifier is simple: if IL-10-biased C9orf72-specific regulatory T cells do not slow progression relative to the controls, the survival-association hypothesis is rejected for that intervention. The choice of the 56–70 epitope follows from the data, since it is both the most broadly recognised and the most broadly HLA-covered, and the choice of mutation carriers follows from their stronger responses, so each design decision traces back to a computed result the reader can check.

The single-cell agent proposes the companion experiment needed to interpret the result. Single-cell transcriptomics with paired T-cell-receptor sequencing on sorted C9orf72-reactive CD4 cells would resolve whether the IL-10 signal comes from a Tr1 or regulatory population, which would support the protective reading, or from a T-helper-2 effector population, which would not, and the receptor sequencing would show whether the response is clonally focused on the dominant

epitopes. These are hypotheses with explicit tests, not claims of efficacy, and they are offered as an illustration of how the grounded outputs can seed a downstream programme, not as a result of this work.

13 Limitations

The system inherits the study's limits and adds its own. FluoroSpot magnitude is in assay-specific spot units with a floor, and the total-response difference comes from the magnitude of responses, while the fraction of donors that respond barely differs. The IFNγ increase does not survive multiple-testing correction. Survival and mutation strata have small per-group samples, so those effects are individually marginal even where consistent with the paper. HLA-II binding uses the authors' NetMHCIIpan predictions and does not re-run the prediction itself. The reasoning agents, in the committed configuration, do not call an external language model, so the generative and critical steps are rule-based and grounded in the tools; the live backend is available but was not used for the recorded results. None of these undermine the reproductions, and each is catalogued with the result it qualifies.

The agentic approach has its own limits that are worth stating plainly. A deterministic generator explores the hypothesis space that its rules and question bank define, so its breadth is bounded by what was anticipated; a live language-model generator would widen that space at the cost of the grounding guarantee, which is the trade this project resolves in favour of grounding. The critique is rule-based and catches the common failure modes, small samples, marginal significance and multiple testing, but it does not reason about confounds it was not told to look for. The therapeutic proposal is a hypothesis assembled from the data and the literature; it is not evidence of efficacy, and the system is built to keep that boundary explicit. These are reasons to read the output as a structured, checkable argument, not as a conclusion.

A mechanistic reading. The study's observations fit a coherent, if still tentative, mechanism. A loss of peripheral tolerance to C9orf72 would expose the protein to autoreactive CD4 T cells, which is consistent with the CD4 dominance of the response and with the stronger responses in mutation carriers, in whom C9orf72 handling is already perturbed. The quality of that response then appears to matter for outcome. An IFNγ-dominated response is the signature of a pro-inflammatory, potentially tissue-damaging T-helper-1 programme, whereas IL-10 is the central regulatory cytokine of suppressive T cells and is linked to neuroprotection through the dampening of microglial activation in several models [9–11]. The finding that the response in ALS is shifted away from IFNγ and toward IL-10, and that the IL-10 share tracks longer predicted survival, is therefore readable as a protective regulatory arm that is present to differing degrees across patients. The decline of regulatory T cells in rapidly progressing ALS [6] places this arm under strain exactly when it is most needed. The intervention that follows from this reading is to reinforce the protective arm, which is why the proposed strategy is tolerogenic and antigen-specific: it targets the C9orf72-specific regulatory response and leaves the rest of the immune system intact.

Why stratification matters here. The data argue for stratification on two axes. Mutation carriers show the strongest C9orf72-specific responses, so they are the subgroup in whom an antigen-specific intervention has the most to act on. Predicted survival separates donors whose response is IL-10-biased from those whose response is not, and the IL-10-biased group is the one with the better outcome, which suggests that the regulatory character of the response, rather than its mere size, is the feature to reinforce. The antigen-level result sharpens this: the IL-10 signal that tracks survival is specific to C9orf72, whereas the IL-5 signal is shared with TDP-43, so an intervention aimed at the protective component should be directed at the C9orf72-specific, IL-10-producing

population, which is narrower than the broader T-helper-2 response. A therapy that expands protective regulatory cells carries a risk in the other direction, that antigen-specific tolerance blunts a response that is itself partly protective, which is why the safety agent keeps this hazard on the list and why the proposed first test is a falsification rather than a dose-finding study.

14 Future work

Three extensions follow naturally, in rough order of value.

The most important is to re-derive the response frequencies from the raw flow-cytometry files. The system already indexes all 90 intracellular-cytokine-staining files and reads the FlowJo workspace gating, so the step is to apply that gating to the event-level data and recompute the cytokine-positive frequencies, then compare them against the authors' summarised values. This would turn the current self-consistency check into an independent quantification, which is the main thing standing between this work and a genuinely external validation. The same event-level data would open the single-cell phenotype to analysis, letting the system ask whether the IL-10-producing C9orf72-reactive cells carry a regulatory or a T-helper-2 surface profile, the question the single-cell agent currently can only propose.

The second is to run the reasoning layer live over the same tools and measure the gap between the deterministic framing and a language-model framing on the grounding and tool-use metrics. That single experiment would convert the project's central design claim, that a grounded core and a reasoning layer can be separated and the trade measured, from an assertion into a result, and it would let the hypothesis generator propose questions beyond the curated bank while the core keeps every reported number honest.

The third is to connect this paper agent to agents built from complementary papers, so a hypothesis generated here can be checked against external datasets and models. Because the outputs are tidy tables, a machine-readable agent registry and JSON records keyed by identifier, and because the tools are reached over a standard protocol, a second agent can consume this one's results without knowing its internals. This is the cross-project ecosystem the programme envisages, and the grounding guarantee is what would keep a chain of such agents trustworthy, where an ungrounded chain would compound guesses.

15 Discussion

15.1 What the system demonstrates

The project set out to test whether an AI system can move from an observation to a falsifiable therapeutic proposal while staying honest about the evidence. The result is a system that reproduces the study from its own data, produces eight verified observations the paper does not state, surfaces two robustness caveats about the original framing, and frames a patient-stratified antigen-specific hypothesis with a companion falsification experiment. The chain from observation to mechanism to hypothesis to computational result to candidate to falsification is present as concrete artefacts, and each link can be inspected on its own. The scientific substance sits in ordinary statistics on real data, which is the part that should be trusted; the agent layer organises the work and decides what to ask.

15.2 Deterministic and language-model reasoning

A central design choice was to separate a deterministic computational core from a reasoning layer. The benefit is that every number in a result is computed, so the grounding rate is one by construction and the discovery loop reproduces bit for bit without credentials. The cost is that the committed run does not explore beyond the hypotheses its question bank encodes. A language-model backend, enabled by an API key, widens that exploration at the price of outputs that must then be checked. The two are not in tension so much as suited to different stages: the language layer is useful for proposing and framing, and the deterministic core is where a proposal becomes a checkable result. The system keeps the boundary visible, which is what lets a reader see where a claim comes from.

15.3 Comparison with the paper

The reproductions agree with the study in direction and, where the authors publish values, in the significance call and to within a small p-value difference. The value the system adds is on top of reproduction. It quantifies the spatial structure of immunogenicity, separates the antigen-specific IL-10 signal from the shared IL-5 signal, and tests the headline effects under corrections the paper does not apply, which both strengthens the robust claims and demotes the IFNγ one. None of this contradicts the paper; it refines what can be said from the same data.

15.4 Generality of the approach

Little in the method is specific to this paper. The parts that are specific are the parser for one workbook layout and the handful of domain tools for epitopes and HLA; everything else, the reproducibility core, the statistics, the tool-registry and MCP layer, the agent framework, the discovery loop and the evaluation, is general. For another data-rich paper the work would be to write a parser for its source data and a small set of domain tools, after which the same scaffolding would provide reproduction, robustness checks, a grounded discovery loop and a self-evaluation. The design therefore reads as a template for turning an open-data paper into a grounded agent, and the cost of applying it falls mostly on the one-off task of understanding and parsing the specific dataset. That is also where the irreducible human judgement sits, because deciding what a column means and whether a reconstructed number matches the source is exactly the verification the author performed here and the part a language model cannot be trusted to do unchecked. The grounding guarantee does not remove the need for that judgement; it localises it, by making every downstream number a mechanical consequence of the parse, so a reader who trusts the parse can trust the rest, and a reader who wants to check the parse has a small, well-defined target. This is the sense in which the contribution is a method and not a single result: the same machinery would carry a different paper, and the measured honesty would transfer with it.

15.5 Ethical considerations

The data are de-identified human immunology measurements, released by the original authors under an open licence for reuse, and this work uses only the derived and summarised values plus the raw files the authors deposited. No attempt is made to re-identify donors. The therapeutic content is the most sensitive output, and it is treated accordingly: the proposals are computational hypotheses with stated falsifiers, not findings, and the report and the public disclaimer both state that they are not medical advice and must be checked against the primary source. The author verified the computational process and the reproducibility of the pipeline, and does not claim to

vouch for the biological validity or clinical relevance of the results, which is the honest position given the subject matter.

15.6 Threats to validity

The threats are easier to see when sorted by type. Construct validity concerns whether the measured quantity means what the claim assumes. The FluoroSpot readout summarises a complex cellular response into a spot count with a floor, so a difference in magnitude is not the same as a difference in whether a response is present, a distinction the censoring analysis makes explicit. Internal validity concerns artefacts in the analysis itself. The overlap of the tiling peptides inflates spatial autocorrelation, which is why the clustering result is read against a permutation null and not as proof of discrete epitopes, and the donor-ordering null control guards against an ordering artefact. External validity concerns whether the findings would generalise. The most important limit here is that the reproductions and the derived observations are built on the authors' own summarised source data and their HLA binding predictions, so the verification is largely a check of self-consistency, not an independent confirmation; the raw flow-cytometry files, which the system indexes but does not yet re-gate, are what would make the quantification independent, and re-deriving the response frequencies from them is the single most valuable next step. Statistical-conclusion validity concerns the inference. Several strata, notably the mutation and survival groups, are small, so individual effects are marginal even where their direction matches the paper, and the predictor is trained on fewer than a hundred peptides, so only the constrained linear model is reported. Each of these is recorded with the result it qualifies.

16 Conclusions

The project shows that a published paper and its open data can be turned into a grounded, interactive agent that does more than summarise. The system reproduces the study's claims from its own data with high fidelity, matching the authors' reported p-values to within $|\Delta p| = 0.00131$, and on top of reproduction it reports two genuinely new biological observations, a set of robustness audits that say which of the paper's effects survive a change of assumption, and one falsifiable downstream hypothesis. Every number is computed, logged and re-verified, and the same tools that answer a question interactively drive the discovery loop, so the interactive agent and the pipeline agree by construction.

The contribution that should outlast the specific findings is the design. By routing every reported quantity through a shared deterministic core and keeping the reasoning layer separate and optional, the system makes its own honesty measurable, which is what the grounding, determinism and false-hypothesis-rejection results demonstrate. That design is also what makes the system safe to extend and to combine with others, because a consumer of its outputs inherits computed, auditable values and not a chain of unverified assertions. The honest limits are stated throughout: the verification is largely self-consistency until the raw flow-cytometry files are re-derived, the small strata make some effects marginal, and the committed discovery is bounded by a curated question bank. None of these undercut the central result, which is that grounding and verification can be built into an AI scientific workflow and then measured, and that doing so costs some ambition at the proposing stage and buys a great deal of trust at the reporting stage.

Reproducibility and AI use. All data fetching is checksum-verified, dependencies are pinned, seeds are fixed, and each run writes a provenance manifest. The AI tools used, the agent specifications and prompts, and the record of hypotheses, evidence and rejected claims are documented in the repository (`AI_USAGE.md`, `agents/registry.json`, `results/discovery/`).

The code is public at https://github.com/samueledelloca/iaiss-project and the derived datasets and results are archived on Zenodo at 10.5281/zenodo.23598390 under CC-BY 4.0. The author inspected and tested all AI-assisted code and is responsible for the scientific content.

A Agent roster

The eighteen agents across the three operational layers are listed in Table 13. Each is defined by a specification with a system prompt and a permitted tool set, exported to `agents/registry.json`, and every run is logged with its prompt, tool calls and output.

Table 13: The agent roster (Layers 3–5).

Agent	Layer	Role
literature agent	Scientific	Find external support/contradiction for findings.
immunology agent	Scientific	Interpret cytokine responses and Th phenotype.
als agent	Scientific	Relate findings to ALS biology, mutation status and contro
hla agent	Scientific	Analyse HLA typing, restriction and binding coverage.
tcell agent	Scientific	Analyse CD4/CD8 and T-cell subset composition.
drug discovery agent	Scientific	Propose antigen-specific therapeutic strategies.
structural agent	Scientific	Analyse epitope sequence/position and modifications.
singlecell agent	Scientific	Propose single-cell follow-ups for phenotype identity.
clinical agent	Scientific	Relate responses to survival and clinical variables.
safety agent	Scientific	Stress-test robustness, multiple testing and censoring.
hypothesis generator	Discovery	Generate testable questions from the data.
hypothesis critic	Discovery	Adversarially critique each hypothesis/answer.
evidence checker	Discovery	Independently recompute and verify each finding.
replication agent	Discovery	Re-run robustness checks (jackknife, FDR, censoring).
contradiction agent	Discovery	Classify each finding vs the paper's claims.
experiment designer	Discovery	Design the most efficient falsification experiment.
falsification agent	Discovery	State what observation would falsify each claim.
pi agent	PI	Integrate all outputs; cannot override contradictory evide

B Contradictions, uncertainties and limitations

The principal-investigator agent compiles the catalogue in Table 14. Items marked as caveats qualify the paper's framing; items marked as limitations or uncertainties qualify the present analyses.

C Reproducibility checklist

Fixed random seeds (`configs/base.yaml`); exact dependency pins (`requirements.txt`, `environment.yml`); data versioned by Zenodo record identifier and per-file MD5; per-run provenance manifests (git commit, configuration hash, package versions, input checksums); deterministic discovery loop verified by re-running; logged agent runs; recorded prompts and agent specifications; and executable stage scripts driven by a single entry point. The AI tools used and the record of hypotheses, evidence and rejected claims are documented in `AI_USAGE.md` and under `results/discovery/`.

Table 14: The catalogue produced by the discovery loop.

ID	Type	Item
Q13	caveat	ALS>HC IL-5 enrichment is significant for: C9orf72, TDP-43. Not unique to C9orf72.
Q18	caveat	Rank-based p=0.00159; above-floor responders 27/28 (ALS) vs 22/28 (HC), Fisher p=0.101: weaker under censoring.
Q19	caveat	After BH-FDR over all 16 antigen x cytokine comparisons, 3/4 C9orf72 cytokines remain significant (q<0.05): Total q=0.0127, IFNg q=0.122, IL
L1	limitation	FluoroSpot magnitude is in assay-specific spot units with a floor of 100; the ALS>HC total-response difference weakens when reduced to the f
L2	limitation	The C9orf72 IFNg ALS>HC increase does not survive BH-FDR across the 16-comparison family (q=0.12), unlike IL-5/IL-10/Total.
L3	uncertainty	Survival and mutation strata have small per-group n; effects are consistent with the paper but individually marginal.

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